Rodrigo Linfati, PhD

Curriculum Vitae (short)

Last update: November 7, 2026

Full Professor - Universidad del Bío-Bío

About Me

Rodrigo Linfati is currently a Full Professor (since 2026) in the Department of Industrial Engineering at the Universidad del Bío-Bío in Concepción (Chile), where he joined in 2012. Dr. Linfati earned his Ph.D. in Operational Research from the University of Bologna (Italy). He also earned a Master of Science in Industrial Engineering, an Industrial Engineering Degree, and a Bachelor of Science in Engineering from the Universidad de Concepción (Chile).

- Research Areas: operations research, combinatorial optimization, metaheuristics, and operations management.
- Research Interests: heuristic and metaheuristic algorithms (TS, SA, VNS, GA, ACO); mathematical modeling and mixed-integer programming formulations; matheuristics and hybrid optimization techniques; machine learning for optimization, including reinforcement learning; quantum and quantum-inspired optimization (metaheuristics, QUBO, quantum annealing, QAOA).
- Application Areas: vehicle routing and last-mile logistics; supply chain management (forestry, sawmill, mining); cutting stock (sawmill, forestry); scheduling, location, and healthcare.

Research Summary

Google Scholar, as of November 7, 2026: Citations: 1545 · h-index: 22 · i10-index: 35.

- Publications: 78 (journal articles: 62; conference articles and book chapters: 16; preprints: 1) - indexes (a publication can appear in more than one index): WoS 53, Scopus 63, Google Scholar 75, SciELO 15
- Theses supervised: 126 (PhD: 2; MSc: 26; Undergraduate: 98)
- Grants as PI: 8 (External funding: 3; UBB funding: 2; Research groups: 3)

Profiles

- ORCID iD: 0000-0002-7659-383X
- Google Scholar profile: 6bJ02OAAAAAJ
- Scopus Author ID: 55250033800
- Web of Science ResearcherID: X-9326-2018
- GitHub: rlinfati

- LinkedIn: rlinfati

Academic Appointments

- **2012-2026+.** Universidad del Bío-Bío, Concepción, Chile - Department of Industrial Engineering
  - Full Professor: 2026+
  - Associate Professor: 2021-2026
  - Assistant Professor "A": 2017-2021
  - Assistant Professor "B": 2014-2017
  - Instructor: 2012-2014

Education

- **2009-2011.** PhD in Automatic Control System and Operational Research / *Dottore di Ricerca in Automatica e Ricerca Operativa* - Alma Mater Studiorum - Università di Bologna, Bologna, Italy
- **2006-2008.** Master of Science in Industrial Engineering - Universidad de Concepción, Concepción, Chile
- **2002-2007.** Industrial Engineering Degree (6 years) - Universidad de Concepción, Concepción, Chile
- **2002-2005.** Bachelor of Science in Engineering - Universidad de Concepción, Concepción, Chile

External Funding

- **2026-2027.** Consolidation of scientific capacities in resilient logistics and supply chain viability under climate change - *PI* - ANID-REC-FOVI-250059 - International Collaboration for Research Institutions - Chile - Team: R. Linfati (UBIOBIO), V. Ortiz (UBIOBIO), C. Rey (UBIOBIO), C. Contreras-Bolton (UDEC), D. Cifuentes (UCSC), R. Soto (ULAGOS), A. Ávila (UFRO), M. González (UFRO), A. Osorio-Mora (RPTU, DE), D. Morillo-Torres (PUJ, CO), J.W. Escobar (UNIVALLE, CO), J.J. Salazar (ULL, ES)
- **2023-2027.** Trajectory-based metaheuristic algorithms for new variants of the VRP - *PI* - ANID-FONDECYT-Regular-1230125 - Fondecyt Regular - Chile - Team: R. Linfati (UBIOBIO), C. Contreras-Bolton (UDEC), C. Rey (UBIOBIO)
- **2016-2018.** Heuristics for Electric Multi-Depot Vehicle Routing Problems and 3D Loading Constraint Problems - *PI* - ANID-FONDECYT-Initiation-11150370 - Fondecyt Initiation in Research - Chile - Team: R. Linfati (UBIOBIO)

Journal Articles - WoS (SCIE, SSCI, AHCI)

1. J.F. Mena-Reyes, R. Soto-Concha, G. Gatica and R. Linfati (2026). Dynamic Generation of Cutting Patterns in Sawmills for Sustainable Planning. *Mathematics*, 14(1), 1-33. doi:10.3390/math14010010
2. R. Soto-Concha, D. Morillo-Torres, J.W. Escobar, J.F. Mena-Reyes and R. Linfati (2025). Mixed-Integer Linear Programming Models for the Vehicle Routing Problem with Release Times and Reloading at Mobile Satellites. *Mathematics*, 13(22), 1-31. doi:10.3390/math13223638

3. J.F. Mena-Reyes, R. Soto-Concha, F. Vergara, V. Ortiz-Araya, J.W. Escobar and R. Linfati (2025). Integrating Social Conflicts into Sustainable Decision-Making of the Forest-to-Lumber Supply Chain. *Forests*, 16(11), 1-26. doi:10.3390/f16111644
4. E. Santana, J.W. Escobar and R. Linfati (2025). A heuristic approach for the last mile delivery with consistent considerations and minimum service for a supply chain. *Mathematics*, 13(10), 1-31. doi:10.3390/math13101553
5. R. Soto-Concha, J.W. Escobar, D. Morillo-Torres and R. Linfati (2025). The Vehicle-Routing Problem with Satellites Utilization: A Systematic Review of the Literature. *Mathematics*, 13(7), 1-28. doi:10.3390/math13071092
6. F.J. Olier, C.A. Porto, G. Herrera-Vidal, W. Adarme-Jaimes, R. Linfati, G. Gatica and J.R. Coronado-Hernández (2025). Allocation of Strategic Positions for Storage of Meat Products Requiring Cold Chain. *Foods*, 14(6), 1-19. doi:10.3390/foods14061010
7. D. Gasset, F. Paillalef, S. Payacán, G. Gatica, G. Herrera-Vidal, R. Linfati and J.R. Coronado-Hernández (2024). Route Optimization for Open Vehicle Routing Problem (OVRP): A Mathematical and Solution Approach. *Applied Sciences*, 14(16), 1-12. doi:10.3390/app14166931
8. F. Muñoz and R. Linfati (2024). Bounding the Price of Anarchy of Weighted Shortest Processing Time Policy on Uniform Parallel Machines. *Mathematics*, 12(14), 1-12. doi:10.3390/math12142223
9. C. Rozas, O. Erazo, V. Ortiz-Araya, R. Linfati and C. Montero (2024). Experimental and Statistical Determination of Convective Heat and Mass Transfer Coefficients for Eucalyptus nitens Sawn Wood Drying. *Forests*, 15(8), 1-12. doi:10.3390/f15081287
10. K. García-Vásquez, R. Linfati and J.W. Escobar (2024). A three-phase algorithm for the pollution traveling Salesman problem. *Heliyon*, 10(9), 1-18. doi:10.1016/j.heliyon.2024.e29958
11. J.F. Mena-Reyes, F. Vergara, R. Linfati and J.W. Escobar (2024). Quantitative Techniques for Sustainable Decision Making in Forest-to-Lumber Supply Chain: A Systematic Review. *Forests*, 15(2), 1-32. doi:10.3390/f15020297
12. Y. Chandía-Rodríguez, R. Linfati, G. Murillo-Vargas and J.W. Escobar (2024). Valuation of Active Chilean Employment Support Policies Seeking Economic Sustainability through Market Flows. *Sustainability*, 16(2), 1-20. doi:10.3390/su16020627
13. C. Cataldo-Díaz, R. Linfati and J.W. Escobar (2024). Mathematical models for the electric vehicle routing problem with time windows considering different aspects of the charging process. *Operational Research*, 24(1), 1-35. doi:10.1007/s12351-023-00806-5
14. M. Núñez-Muñoz, R. Linfati and J.W. Escobar (2023). Two-stage optimization scheme of routing scheduling from a single distribution center to multiple customers. *Operational Research*, 23(2), 1-29. doi:10.1007/s12351-023-00747-z
15. F. Corral, E. Forcael and R. Linfati (2023). Workforce Scheduling Efficiency Assessment in Construction Projects Through a Multi-Objective Optimization Model in the Covid-19 Context. *Heliyon*, 9(6), 1-16. doi:10.1016/j.heliyon.2023.e16745
16. L. Aburto, R. Romero-Romero, R. Linfati and J.W. Escobar (2023). An Approach for a Multi-Period Portfolio Selection Problem by considering Transaction Costs and Prediction on the Stock Market. *Complexity*, 2023, 1-15. doi:10.1155/2023/3056411

17. C. Rojas, R. Linfati, R. Scherer and L. Pradenas-Rojas (2023). Using Geopandas for locating virtual stations in a Free-floating Bike-sharing System. *Heliyon*, 9(1), 1-14. doi:10.1016/j.heliyon.2022.e12749
18. H. Barreto, J.W. Escobar, R. Linfati and V. Ortiz-Araya (2022). Disciplinary Categorization of the Cattle Supply Chain - A review and bibliometric analysis. *Sustainability*, 14(21), 1-44. doi:10.3390/su142114275
19. O. Coronado-Hernández, D. Bonilla-Correa, A. Lovo, V. Fuertes-Miquel, G. Gatica, R. Linfati and J.R. Coronado-Hernández (2022). An Implicit Formulation for Calculating Final Conditions in Drainage Maneuvers in Pressurized Water Installations. *Water*, 14(21), 1-14. doi:10.3390/w14213364
20. R. Linfati, F. Yáñez-Concha and J.W. Escobar (2022). Mathematical models for the Vehicle Routing Problem by considering Balancing Load and Customer Compactness. *Sustainability*, 14(19), 1-20. doi:10.3390/su141912937
21. J. Echeverri-Díaz, O. Coronado-Hernández, G. Gatica, R. Linfati, R. Méndez-Anillo and J.R. Coronado-Hernández (2022). Sensitivity of Empirical Equation Parameters for the Calculation of Time of Concentration in Urbanized Watersheds. *Water*, 14(18), 1-24. doi:10.3390/w14182847
22. C. Flores-Fonseca, R. Linfati and J.W. Escobar (2022). Exact algorithms for production planning in mining considering the use of stockpiles and sequencing of power shovels in open-pit mines. *Operational Research*, 22(3), 2529-2553. doi:10.1007/s12351-020-00618-x
23. J. Salgado-Cassiani, O. Coronado-Hernández, G. Gatica, R. Linfati and J.R. Coronado-Hernández (2022). Probabilistic Approach to Determine the Spatial Distribution of the Antecedent Moisture Conditions for Different Return Periods in the Atlántico Region, Colombia. *Water*, 14(8), 1-24. doi:10.3390/w14081217
24. C. Cataldo-Díaz, R. Linfati and J.W. Escobar (2022). Mathematical model for the electric vehicle routing problem considering the state of charge of the batteries. *Sustainability*, 14(3), 1-26. doi:10.3390/su14031645
25. B. Valenzuela-Fonseca, R. Linfati and J.W. Escobar (2022). Strategy for locating people to reduce the transmission of COVID-19 using different interference measures. *Sustainability*, 14(1), 1-19. doi:10.3390/su14010529